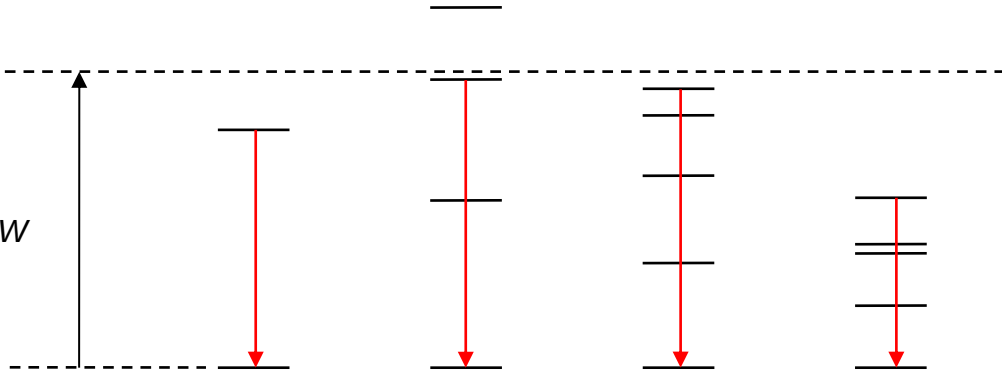


27	B To achieve shortest wavelength, this implies that the photon emitted has the highest frequency. Since energy of a photon is hf and is equal to the energy difference between two levels, the highest possible transition possible for each atom is shown in the diagram below. Hence atom B will give the greatest energy difference.

	 The diagram illustrates four vertical red arrows, labeled A, B, C, and D, each with horizontal bars at the top and bottom. A dashed line is at the top, and a vertical arrow labeled W indicates the distance from the bottom dashed line to the top dashed line. A B C D
28	C